

September 2026

AI-Powered Fraud Detection

Fraud Detection Initiative

Executive Presentation

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Executive Summary

Manual review of transactions and one-size-fits all tagging of fraudulent transactions cannot catch-up on the growing number of volume of digital and mobile-banking transactions.

Problem Statement

We barely catch fraud transactions

0.19%

of actual fraud cases caught by today's static and rule-based system

(16 out of 8,213 fraud cases in our evaluation sample)



Fraud losses hit the financials and are unrecovered



Widening gap between manual review and growing number of transactions

Proposed Solution

Leverage model that flags a transaction before an actual fraud payout happens

Using three AI approaches on historical transaction data, the best model was determined to catch a fraudulent transaction. It learns a blueprint of a digital transaction and flag a transaction for a higher risk of fraud. This flag routes the transaction for review before an actual payout happens.

1

Learn

Train using historical data to determine which transaction is legitimate vs fraudulent

2

Score

There is a real-time risk score on correct tagging of fraudulent transaction

3

Flag

High-risk transactions to be flagged and passed on to human reviewers before transaction gets completed

Pilot Evaluation Results

99.6%

of fraudulent
transactions caught

0

Legitimate transactions
flagged as fraud

99.6% vs 0.19% result of existing rule-based tagging of fraudulent transactions

Evaluation results are based on the existing sample data of digital / mobile-banking transactions with tagging of legitimate and fraudulent transactions

Key Assumptions and Constraints

Fraud detection markers

Key hints that may result to a fraudulent transaction was based on the sample data. These markers may or may not be present on live-data so results may be lower than the previous slide.

Side-by-side run

For implementation, current rule-based tagging and model can be run in parallel to assess performance.

Key Risk/Finding

30% of fraud caught for lower-balance customer vs 100% for other segments

Precision is at 1.0 across customer balance segments; however, it might miss fraud against the lower-balance customer more often. This is a potential risk and gap into fully developing a fraud detection initiative.



Implementation Plan

- 1 Secure approval, build steering committee**
Secure the necessary approval and funding to proceed; build working group and project timings
- 2 Fix customer segmentation gap**
Implement fix on customer segment for low-balance customers
- 3 Implement parallel run**
Leverage two models to utilize existing rule-based systems and new model on live data
- 4 Evaluate and rerun**
Assess results and fix necessary model opportunities
- 5 Phased implementation**
Deploy alongside human review, expand as confidence level builds

Key Business Benefit

Business value = *Fraud amount prevented*
- *cost of reviewing false flagging* - *cost of fraud still missed*

Key wins from sample dataset

- 2,455 out of 2,464 fraud cases caught
- No legitimate transactions flagged

What's next to complete business value?

- Cost of manual review per transaction
- Recovery value of fraud prevented



Risks and Mitigation Plan

Inflated results

Due to sample dataset used, utilize parallel run on existing rule-based system and new model to assess results

Synthetic training data

Model trained on simulated data; leverage live data prior to implementation

Customer segmentation gap

Technical fix prior to implementation





What's Next –

- Executive approval to initiate project and deploy parallel run

- Working group and independent partner for risk, audit and compliance

- Project governance and check-in to report results